

Sana Nazir

snazir@purdue.edu | sananazir.github.io

PhD researcher developing high-order, hardware-agnostic, performance-portable CFD solvers in C++/Kokkos.

EDUCATION

PHD AERONAUTICS & ASTRONAUTICS
Purdue University
—Expected Dec 2027—

MSC AERONAUTICS & ASTRONAUTICS
Purdue University

MSC COMPUTER SCIENCE
Purdue University

BSC COMPUTER SCIENCE
Lahore University of
Management Sciences

CONCENTRATIONS

Graduate Coursework

- Aerodynamics
- Propulsion
- Networks & OS
- Compilers

ACADEMIC PROJECTS

- Lab tool for visualizing visible emission spectra of noble-gas plasmas based on T_e (Python)
- Identification of Vortex Stretching Events in Turbulent Channel Flow (JHTDB)
- Hypersonic DSMC Study of Hollow-Cylinder-Flare (SPARTA)
- Hydrogen-Powered Micro-Gas-Turbine
- Ultra-High-Bypass-Ratio Geared Turbofan Engine Analysis
- MapReduce Word Count (C++, OpenMP, MPI)
- GCC Compiler Support for Analyzing Uninitialized Local/Global Variables
- XINU OS Components Including Scheduler & Virtual Memory Support
- UDP-Based Chat App
- UDP-Based Audio Streaming App

EXPERIENCE

PURDUE UNIVERSITY | PHD STUDENT | GRADUATE STAFF

School of Aeronautics & Astronautics

- Jan 2023 - present | Teaching/Research Assistant, Lab Instructor

INTEL | SYSTEMS/ COMPILER ENGINEER

- Aug 2019 - July 2021 | Software Engineer IOTG - Compiler Architecture
- Oct 2018 - Aug 2019 | Software Engineer IAGS - Systems Software
- May 2017 - Aug 2017 | Graduate Software Engineering Intern
- May 2016 - Aug 2016 | Graduate Software Engineering Intern

SELECT PROJECTS

PURDUE UNIVERSITY

HIGH-ORDER CFD FRAMEWORK IN C++/KOKKOS

Developing a hardware-agnostic high-order CFD framework using explicit and compact finite-difference schemes on structured and curvilinear meshes, targeting performance portability across CPU/GPU architectures with OpenMP, CUDA, and HIP.

- Accepted @ AIAA SCITECH 2027
- Presented @ HPSF Conference 2026; AIAA SCITECH 2026.

AEROSPACE SCIENCES LAB | SBLI AT MACH 2.0

Developing a color schlieren visualization technique for experimental investigation of swept oblique shock-wave/boundary-layer interactions (SBLI) at Mach 2.0.

- Presented @ AIAA SCITECH 2025.

PLASMA & ELECTRIC DISCHARGES COURSEBOOK

Co-authoring a comprehensive graduate-level textbook on Plasmas and Electric Discharges.

INTEL

PLAIDML EXECUTION PROVIDER FOR ONNX RUNTIME

Created the interface to enable ONNX Runtime with PlaidML as the execution provider, by extending the Tile eDSL to include ONNX operations.

MOHAWK RIVER AMBIENT COMPUTE PROTOTYPE

Built a Face Recognition Module utilizing a 360 camera device for a dual-OS shared peripheral system.

- Prototype showcased @ COMPUTEX 2019, INTEL DTTC 2019, INTEL SWPC 2019, INTEL Demo Day 2020

FEDERATED LEARNING DEMONSTRATION

Demonstrated Federated Learning using a two-layer CNN on the MNIST dataset. Setup with Intel NUC devices and multiple connected smartphones. Built using TensorFlow and Python.

- Presented as live demo and poster @ Intel "Showcase Your Amazing" 2017.

TELECONFERENCING & IROBOT OBSTACLE AVOIDANCE

Developed a 360° teleconferencing application with automatic speaker detection and an iRobot obstacle-avoidance system using Intel RealSense depth cameras and computer vision.

- Prototype showcased at Intel DTTC 2017.
- Significant Effort & Achievement Award for measurable work-group impact.